

Phase1: Defect Exploration

Phase1 System Prompt

- **Goal:** Locate and understand the bug through code exploration writing tests
- **Output:** Produce a structured summary before transitioning to test writing

Execute

Explore

Isolated Exploration Tools

Search Code

Search on the repository graph

Read File

Get the detailed file content

Ready to Test

Transition to test writing phase

Exploration Completion Check

Have I gathered enough information to clearly understand the root cause and impacted behavior?

Trigger Tool Call

Invoke the tool to generate a structured exploration summary.

Issue

Incorrect parenthesizing of Subs Here is an example.
```python  
...  
```

Code Base

sympy
printing
tests
test_latex.py
latex.py
...

Test

Protocol

Shared message Shared Context

Bug Loc.
File/Func/Line

Root Cause
Logic error

Exp. Vs Act.
Expect/Observe

Reproduction
Input/Trigger

Ref Tests
Style/Convention

Test Plan
Planned Tests

Phase2: Test Generation

Phase2 System Prompt

- **Goal:** Write unit tests that reproduce the bug without fixing it
- **Output:** A verified test file that fails on buggy code and passes after the fix

Execute

Isolated Testing Tools

Edit File

Edit or create file for test case

Run Tests

Execute generated test with python

Submit

Submit the test for evaluation

Test Selection

Candidate Tests

Issue and rewrites

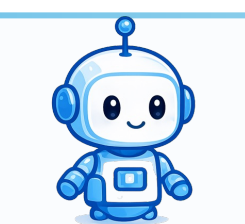
Select

Candidate Patches

Surrogate code patches

Select Best Test

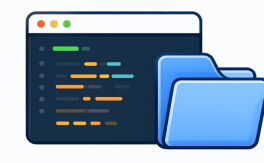
Select the test with the highest robust score.



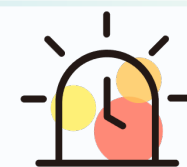
Explorer



Exploration System Prompt



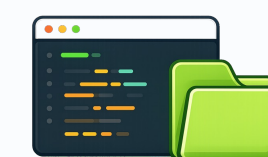
Explore on Code Base



Summary
from exploration to testing



Testing System Prompt



Writing Test for Reproduction



Tester